

AI ASSISTED CODING

LAB - 7.5

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BATCH-12

Task 1 (Mutable Default Argument – Function Bug)

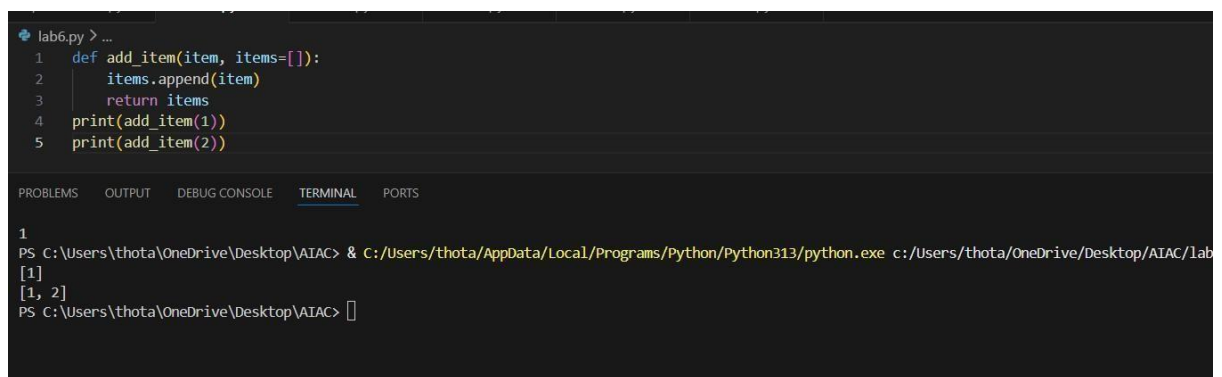
Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it. # Bug: Mutable default argument

```
def add_item(item, items=[]):
```

```
    items.append(item)
    return items
```

```
print(add_item(1))
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug.



```
lab6.py > ...
1 def add_item(item, items=[]):
2     items.append(item)
3     return items
4 print(add_item(1))
5 print(add_item(2))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
1
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
[1]
[1, 2]
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

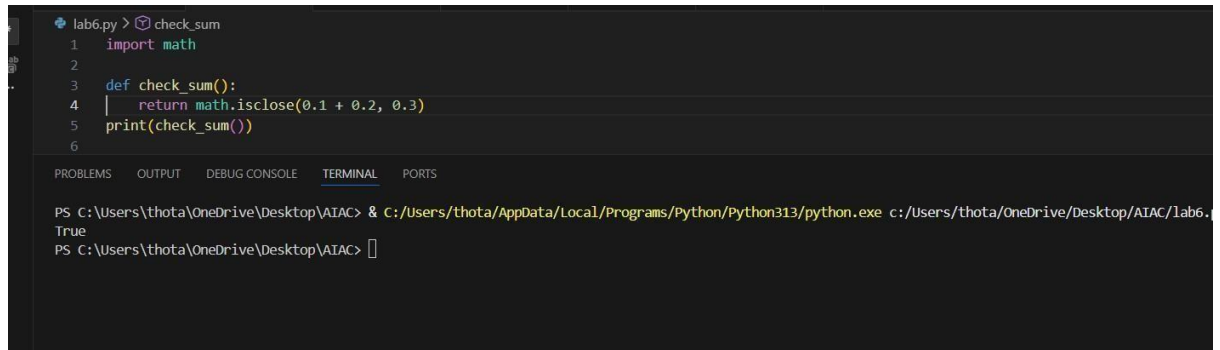
Task 2 (Floating-Point Precision Error)

Task: Analyze given code where floating-point comparison fails. Use AI to correct with tolerance. # Bug: Floating point precision issue

```
def check_sum(): return (0.1 + 0.2)
```

```
== 0.3 print(check_sum())
```

Expected Output: Corrected function



```
lab6.py > check_sum
1 import math
2
3 def check_sum():
4     return math.isclose(0.1 + 0.2, 0.3)
5 print(check_sum())
6

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

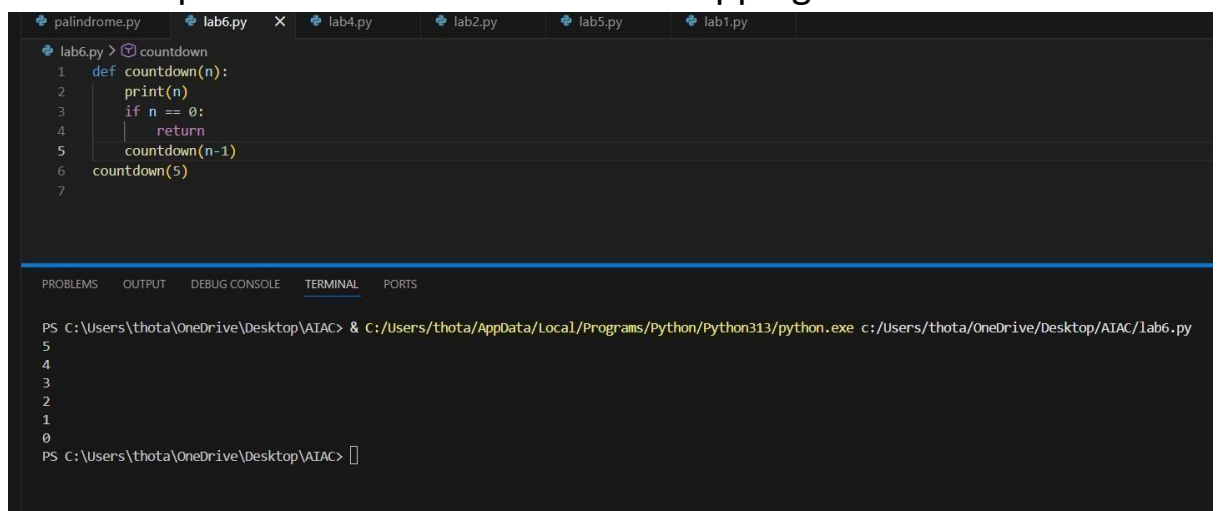
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
True
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

Task 3 (Recursion Error – Missing Base Case)

Task: Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix. # Bug: No base case

```
def countdown(n): print(n) return countdown(n-1) countdown(5)
```

Expected Output : Correct recursion with stopping condition



```
palindrome.py lab6.py X lab4.py lab2.py lab5.py lab1.py
lab6.py > countdown
1 def countdown(n):
2     print(n)
3     if n == 0:
4         return
5     countdown(n-1)
6 countdown(5)
7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
5
4
3
2
1
0
PS C:\Users\thota\OneDrive\Desktop\AIAC>
```

Task 4 (Dictionary Key Error)

Task: Analyze given code where a missing dictionary key causes error.

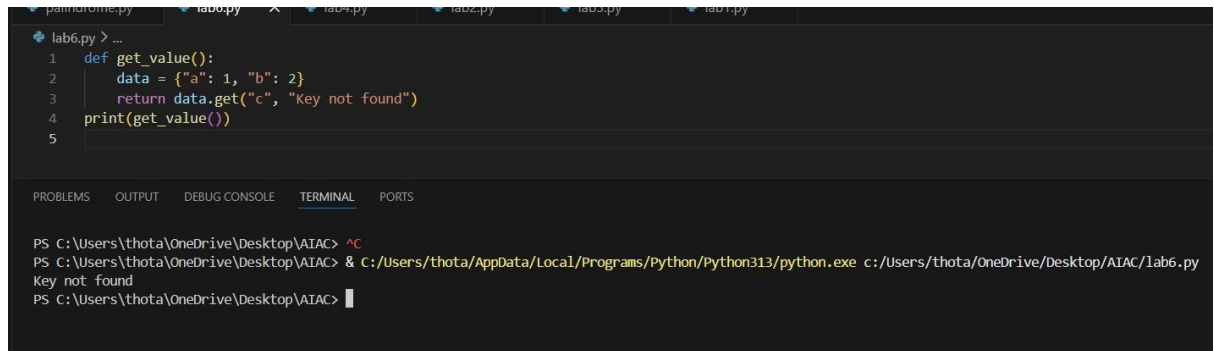
Use AI to fix it.

Bug: Accessing non-existing key def

get_value(): data = {"a": 1, "b":

2} return data["c"] print(get_value())

Expected Output: Corrected with .get() or error handling.

The screenshot shows a code editor with a file named 'lab6.py'. The code defines a function 'get_value()' that creates a dictionary 'data' with keys 'a' and 'b', and then attempts to return 'data["c"]'. This causes a 'KeyError' because 'c' is not a key in the dictionary. The code is then corrected to use 'data.get("c", "Key not found")'. Below the editor, a terminal window shows the command to run the script, followed by the output 'Key not found'.

Task 5 (Infinite Loop – Wrong Condition)

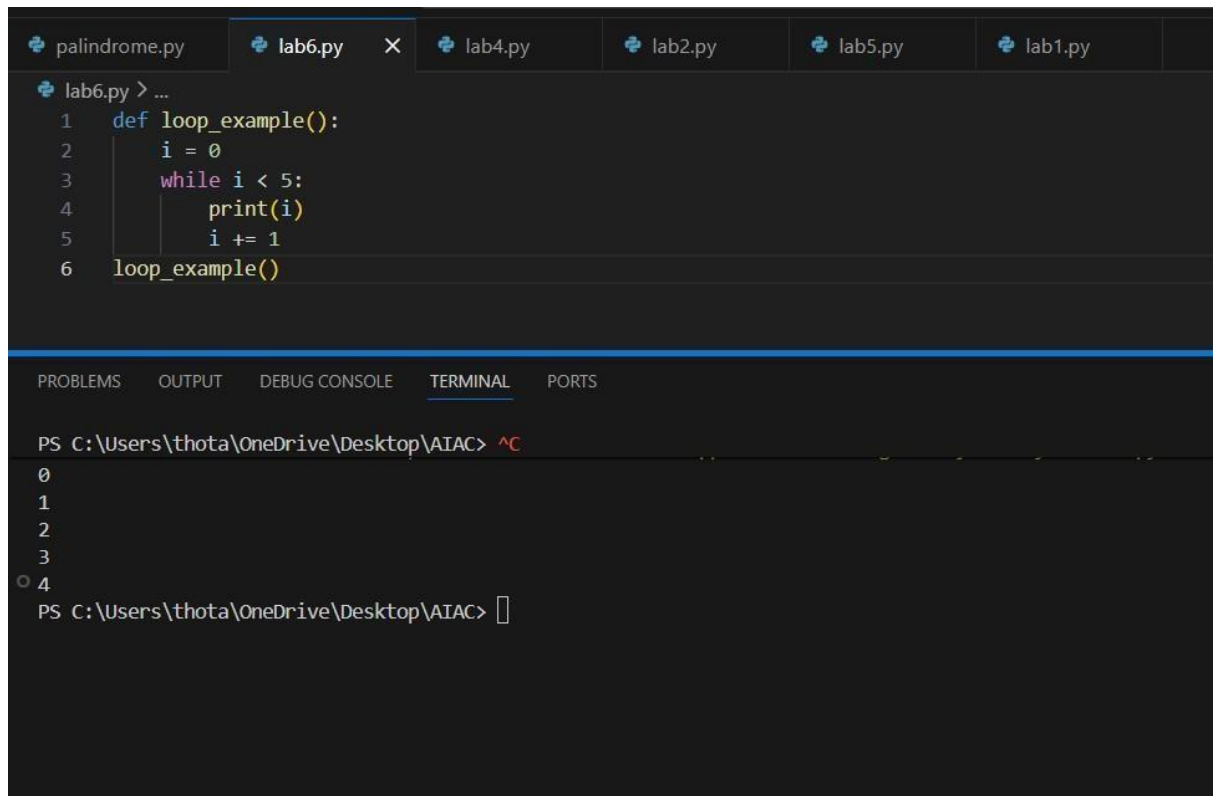
Task: Analyze given code where loop never ends. Use AI to detect and fix it.

Bug: Infinite loop def loop_example():

i = 0 while i

< 5: print(i)

Expected Output: Corrected loop increments i.



The screenshot shows a code editor with several tabs: `palindrome.py`, `lab6.py` (active), `lab4.py`, `lab2.py`, `lab5.py`, and `lab1.py`. The `lab6.py` tab contains the following Python code:

```
1 def loop_example():
2     i = 0
3     while i < 5:
4         print(i)
5         i += 1
6 loop_example()
```

Below the code editor is a terminal window with tabs: `PROBLEMS`, `OUTPUT`, `DEBUG CONSOLE`, `TERMINAL` (active), and `PORTS`. The terminal shows the command prompt output:

```
PS C:\Users\thota\OneDrive\Desktop\AIAC> ^C
0
1
2
3
4
PS C:\Users\thota\OneDrive\Desktop\AIAC> 
```

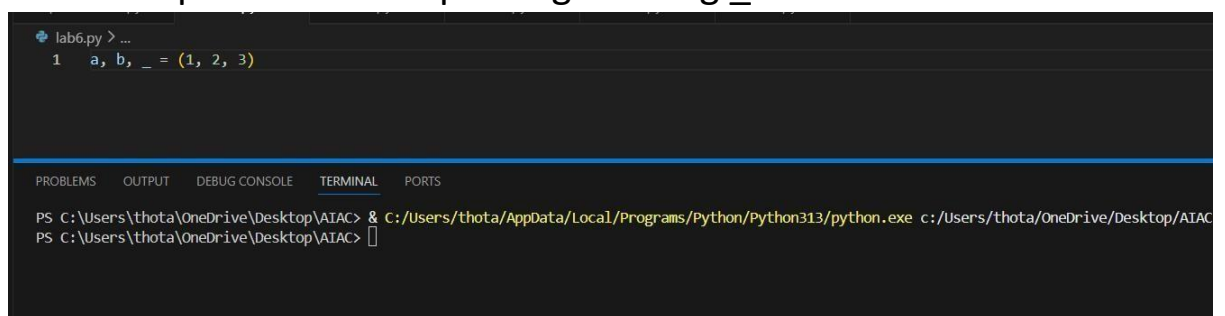
Task 6 (Unpacking Error – Wrong Variables)

Task: Analyze given code where tuple unpacking fails. Use AI to fix it.

Bug: Wrong unpacking

`a, b = (1, 2, 3)`

Expected Output: Correct unpacking or using `_` for extra values.



The screenshot shows a code editor with a single tab: `lab6.py`. The code contains the following line:

```
1 a, b, _ = (1, 2, 3)
```

Below the code editor is a terminal window with tabs: `PROBLEMS`, `OUTPUT`, `DEBUG CONSOLE`, `TERMINAL` (active), and `PORTS`. The terminal shows the command prompt output:

```
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC
PS C:\Users\thota\OneDrive\Desktop\AIAC> 
```

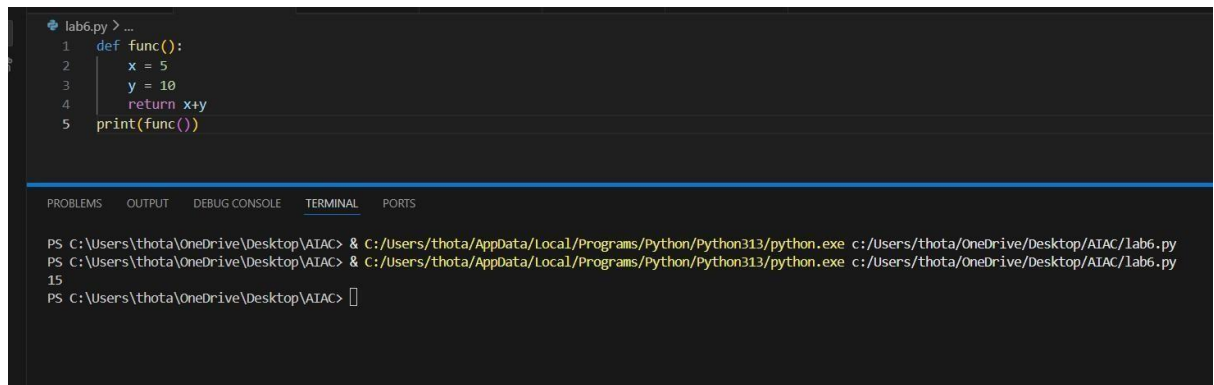
Task 7 (Mixed Indentation – Tabs vs Spaces)

Task: Analyze given code where mixed indentation breaks

execution. Use AI to fix it. # Bug: Mixed indentation def

func(): x = 5 y = 10 return x+y

Expected Output : Consistent indentation applied.



```
lab6.py > ...
1 def func():
2     x = 5
3     y = 10
4     return x+y
5 print(func())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
15
PS C:\Users\thota\OneDrive\Desktop\AIAC> 
```

Task 8 (Import Error – Wrong Module Usage)

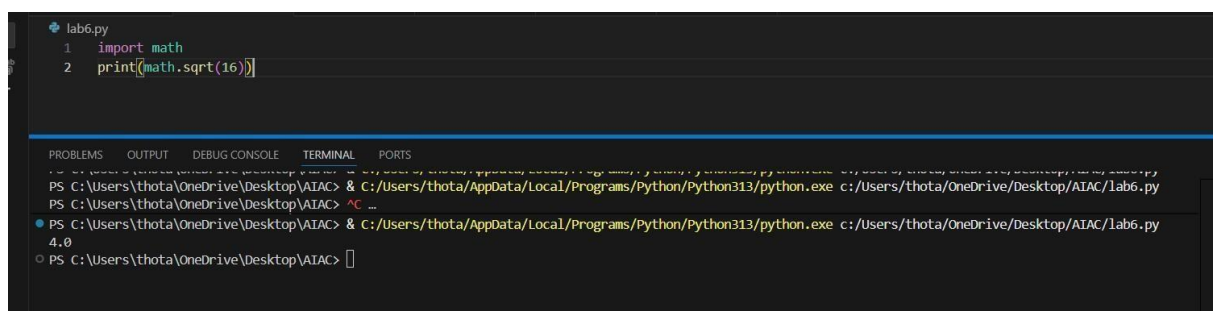
Task: Analyze given code with incorrect import. Use AI to fix.

Bug: Wrong import import

maths print(maths.sqrt(16))

Expected Output: Corrected to

import math



```
lab6.py
1 import maths
2 print(maths.sqrt(16))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
PS C:\Users\thota\OneDrive\Desktop\AIAC> ^C _
PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:/Users/thota/AppData/Local/Programs/Python/Python313/python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab6.py
4.0
PS C:\Users\thota\OneDrive\Desktop\AIAC> 
```